

$$U \sim N_6(0.3\mathbf{1}, \Sigma_X), \quad (\Sigma_X)_{jk} = 0.6e^{-0.3|j-k|}, \quad B_j = \mathbb{I}(U_{j+2} > c_j), \quad c = (0, 0.3, -0.2, 0.1); \quad Y_0 | U \sim N(0.5 + 0.3X_1 + 0.2X_2 + 0.2B_1 + 0.15B_2 + 0.10B_3 + 0.15B_4, 4).$$